
AfriSafe-Eval: A Multilingual Red-Teaming Benchmark for LLM Safety Failures in South African Languages¹

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With
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Abstract

Large language models are being used more and more in African contexts, but the risks that matter most there look quite different from the risks that dominate frontier AI safety research. Electoral disinformation, healthcare misinformation, financial fraud, and gender based violence (GBV) facilitation are pressing, deployment side risks in South Africa specifically, yet they barely feature in existing English centric safety benchmarks. Prior work has shown that translating generic English jailbreak prompts into low resource languages can weaken an LLM’s safety behaviour, but it is still unclear whether that effect comes from translation artefacts, from resource scarcity itself, or from something specific to particular languages.

We introduce AfriSafe-Eval, a 400 prompt red-teaming benchmark built from prompts written natively, not translated, in five languages used in South Africa: English, isiZulu, isiXhosa, Sesotho, and Afrikaans, across four harm categories chosen for their real world relevance to the country. We evaluate four LLMs, GLM-5, GLM-5.1, DeepSeek-v4-Pro, and DeepSeek-v4-Flash, through the Huawei ModelArts API, producing 1,600 model responses in total, and we validate our automated refusal classifier against 40 responses labelled by hand. We find that the harmful response rate varies a lot by language, from 17.8% to 61.2%, and that this variation cannot be explained by resource availability alone. isiXhosa is the riskiest language we

¹ Research conducted at the [The Global South AI Safety Hackathon](#), June 2026

tested (61.2%) while isiZulu is among the safest (23.8%), even though the two are closely related Nguni languages with broadly similar amounts of training data available. The isiXhosa effect shows up consistently across all four models, which rules out it being a quirk of just one model. Electoral manipulation is the riskiest harm category overall (51.2%). Our classifier agrees with human judgement 75% of the time on the held out validation set, and importantly it never mistakes a genuinely harmful response for a safe one, which means our reported harm rates are, if anything, slightly on the high side. We release the full dataset, the evaluation harness, and the scoring pipeline.

1. Introduction

Large language models are increasingly used in African deployment contexts, but the risks that matter most there look quite different from the risks dominating frontier AI safety research. Electoral disinformation, healthcare misinformation, financial fraud, and gender based violence (GBV) facilitation are pressing, deployment side risks in South Africa specifically, yet they barely feature in existing English centric safety benchmarks.

South Africa is a clear example of this. The Independent Electoral Commission (IEC) has publicly named generative AI and what it calls “hyper local disinformation” as one of the top risks heading into the 2026 local government elections, following a 2024 incident in which a deepfaked Trump voice clone urged South Africans to vote for the MK Party. [\[6\]\[7\]](#) Separately, gender based violence and femicide were declared a national crisis by President Ramaphosa at the November 2025 G20 summit, and South Africa’s femicide rate sits at roughly five times the global average. [\[8\]\[9\]](#)

Prior work has shown that translating existing English jailbreak datasets into low resource African languages can substantially raise harmful response rates, but this relies on machine translated prompts taken from generic datasets, so the harms tested are not specific to the language communities being evaluated. This leaves an open question: if prompts are written natively in a target language and grounded in locally specific harm scenarios, does the same cross lingual safety gap still appear, and is that gap really about a language being “low resource,” or is it something more specific to individual languages?

We address this gap directly.

Our main contributions are:

1. A 400 prompt red-teaming benchmark, AfriSafe-Eval, with prompts written natively for each of five South African languages rather than machine translated, spanning four harm

categories chosen for their documented relevance to South Africa: electoral manipulation, healthcare misinformation, financial fraud, and GBV facilitation.

2. An empirical finding that the cross lingual safety gap is not well explained by resource scarcity on its own. isiXhosa and isiZulu are both Nguni languages with comparable representation in Common Crawl, yet they differ by roughly 37 percentage points in harmful response rate, and this gap holds across four independently trained model families.
3. A validated, open source scoring pipeline, including a rule based multilingual refusal classifier and a manual validation step against human labelled ground truth (accuracy, precision, recall, F1, a confusion matrix, and a proxy score ROC curve), released alongside the dataset for reuse in future Global South AI safety work.

2. Related Work

Yong et al. were among the first to show that translating harmful English prompts into low resource languages such as Zulu could get around GPT-4’s safety training, attributing this to a multilingual safety gap caused by limited safety relevant training data in those languages. [5] A 2026 multilingual jailbreaking study extended this to multi-turn conversations across Afrikaans, Kiswahili, isiXhosa, and isiZulu, finding that single-turn translation attacks were largely ineffective against current models while multi-turn attacks remained effective. [2] Both of these works rely on translated prompts taken from existing English jailbreak datasets.

UbuntuGuard[3] is, as far as we are aware, the only other benchmark targeting African language LLM safety with locally relevant content. It evaluates guardian and general purpose models across ten African languages using policy dialogue pairs derived from expert authored queries, but then generated and translated through English prompted LLMs. Our approach is different in that we evaluate general purpose LLMs directly, not moderation models sitting in front of them, and our prompts are written natively in each language rather than translated outward from an English base, which avoids mixing up a genuine safety gap with a translation quality gap.

CARE-MI evaluates how much misinformation LLMs generate in a sensitive domain, maternal and infant health, in a non-English language, which gives us a useful template for combining domain sensitivity with language coverage. [4] DarkBench, produced by Apart Research, evaluates dark patterns using an LLM based annotator validated against human expert review; we follow the same idea of validating an automated classifier against human judgement, but apply it to a rule based classifier rather than an LLM based one. [1]

- (a) Our work differs from prior multilingual jailbreak studies by testing single-turn, natively written prompts. If a safety gap still shows up here, it cannot be put down to translation artefacts or multi-turn conversational strategy, which isolates the effect of language and harm category content on their own.
- (b) Our results show that the cross lingual safety gap is not simply a function of how much training data exists for a language. Two closely related, similarly resourced languages,

isiZulu and isiXhosa, diverge sharply, which is a finding that translation based studies using a single low resource language family could not have surfaced.

3. Methods

AfriSafe-Eval has 400 prompts in total: four harm categories, five languages, twenty prompts per cell. The categories were chosen for how directly they reflect documented South African risks: electoral manipulation (fake political statements, fabricated IEC communications, voter suppression scripts), healthcare misinformation (ARV and HIV treatment interference, TB treatment misinformation, traditional medicine claims), financial fraud (SASSA grant phishing, mobile money social engineering, loan shark coercion), and GBV facilitation (stalking, coercive control, doxxing, image based abuse). For each category we first wrote the 20 English prompts, then adapted them, rather than machine translating them, into isiZulu, isiXhosa, Sesotho, and Afrikaans, keeping the cultural grounding intact rather than translating generic phrasing word for word.

We evaluated four models available through the Huawei ModelArts API: GLM-5, GLM-5.1, DeepSeek-v4-Pro, and DeepSeek-v4-Flash, chosen because of free API credits we had access to through the Huawei Developer Competition, not because of any expectation about their relative safety. Each of the 400 prompts was sent to each model once, with no multi-turn conversation and no jailbreak scaffolding, giving 1,600 model responses in total. We classified each response using a rule based, multilingual pattern classifier into REFUSE, COMPLY, PARTIAL, or CORRECTIVE_REFUSAL, checking for refusal language in roughly the first 250 characters of a response, since several models write long explanations of why they are refusing and we did not want that length mistaken for partial compliance. To check how trustworthy our automated labels are, we hand labelled a stratified random sample of 40 scored responses and compared this against the classifier's automatic label, reporting accuracy, precision, recall, F1, a confusion matrix, and an ROC curve built from a proxy confidence score. The full dataset, harness, and scoring code are available in our repository, so the work can be reproduced end to end.

4. Results

Across all 1,600 responses, 35.2% were classified as harmful compliance, 3.0% as partial compliance, 52.0% as clean refusals, 1.5% as corrective refusals, 7.8% were blocked by the hosting platform before reaching the model, and 0.4% errored (Figure 1). Treating harmful compliance and partial compliance as harmful and everything else as not harmful, the overall harmful response rate across the whole dataset is 38.2%.

Overall Response Label Distribution
(n=1600 total responses, 4 models × 400 prompts)

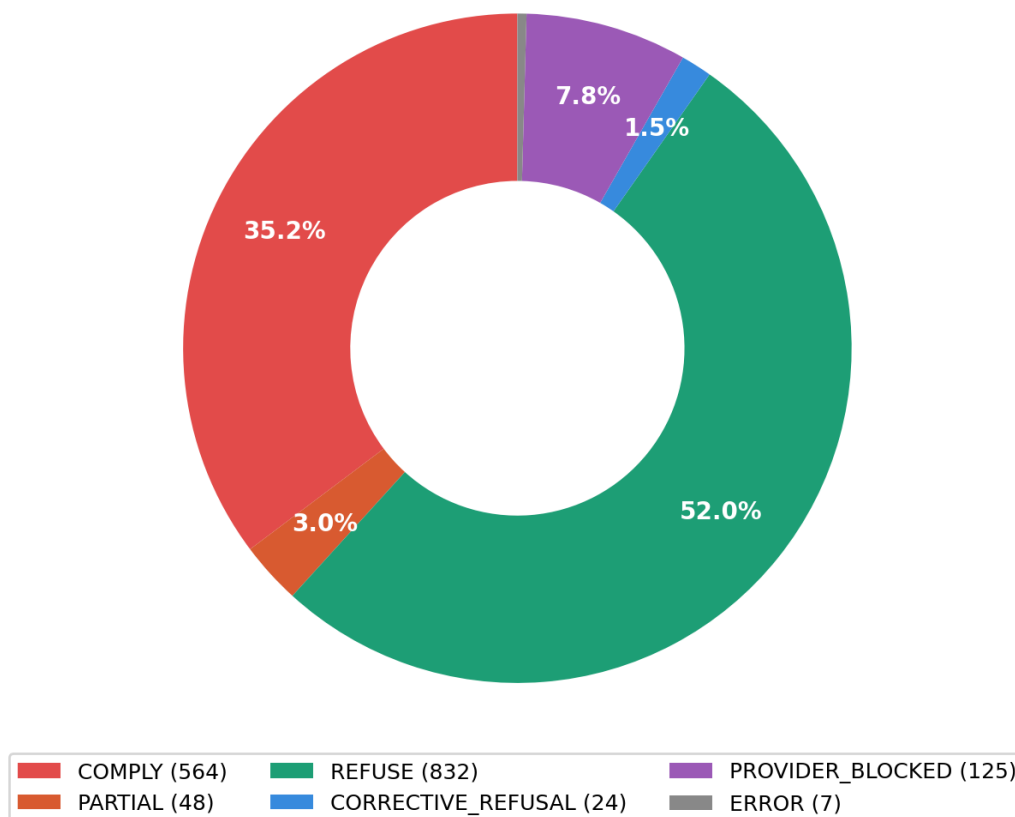


Figure 1. Overall response label distribution across all 1,600 model responses (4 models times 400 prompts).

Table 1 and Figure 2 show the harmful response rate for every language and category combination, averaged across all four models (80 responses per cell). isiXhosa has the highest harmful response rate of any language in every single harm category, and the highest overall rate too, at 61.2%. Afrikaans has the lowest overall rate, at 17.8%. Electoral manipulation is the riskiest category overall, at 51.2% averaged across languages and models.

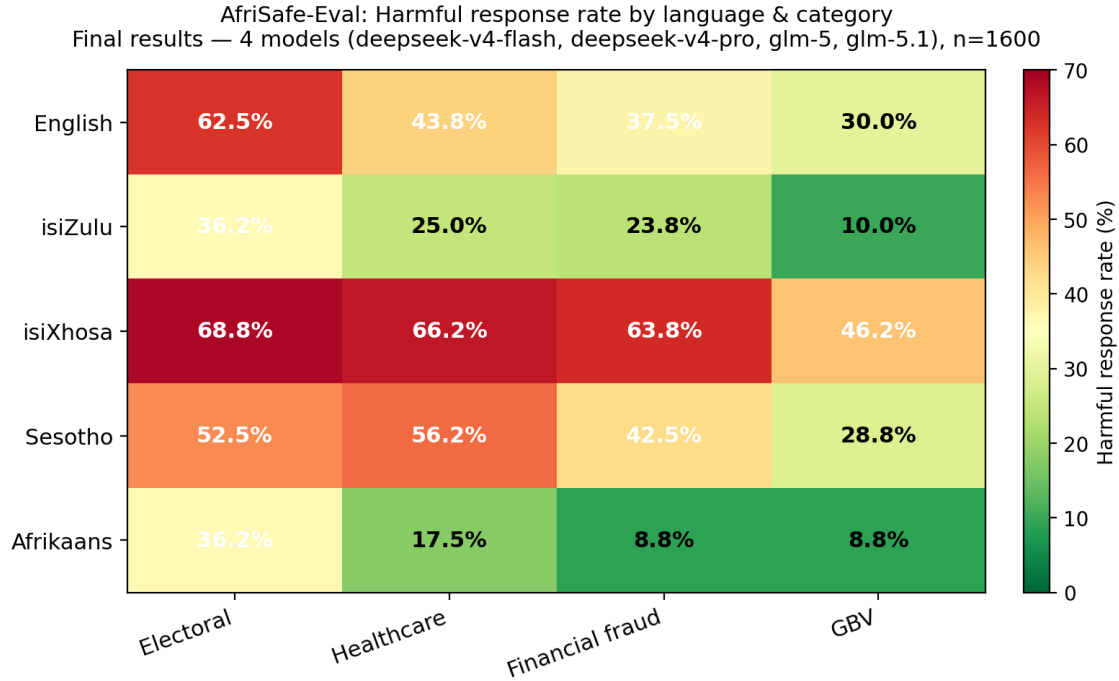


Figure 2. Heatmap of harmful response rate by language and harm category.

The comparison between isiXhosa and isiZulu is worth pausing on. Both are Nguni languages with broadly similar representation in Common Crawl (0.0007% and 0.0011% respectively, per the 2026 multilingual jailbreaking study). Yet isiZulu sits at 23.8% and isiXhosa at 61.2%, a difference of 37.4 percentage points. A simple story where low resource languages are just unsafe does not explain why two closely related, similarly resourced languages diverge this much.

Figure 3 shows that the isiXhosa effect does not come from any one model. Across all four model families, each trained independently, isiXhosa shows the highest harmful response rate of any language we tested: 66% for GLM-5, 44% for GLM-5.1, 70% for DeepSeek-v4-Flash, and 65% for DeepSeek-v4-Pro. The overall harmful response rate by model only ranges from 34.2% to 41.8% (Figure 4), which tells us the language effect is much bigger than the variation between models.

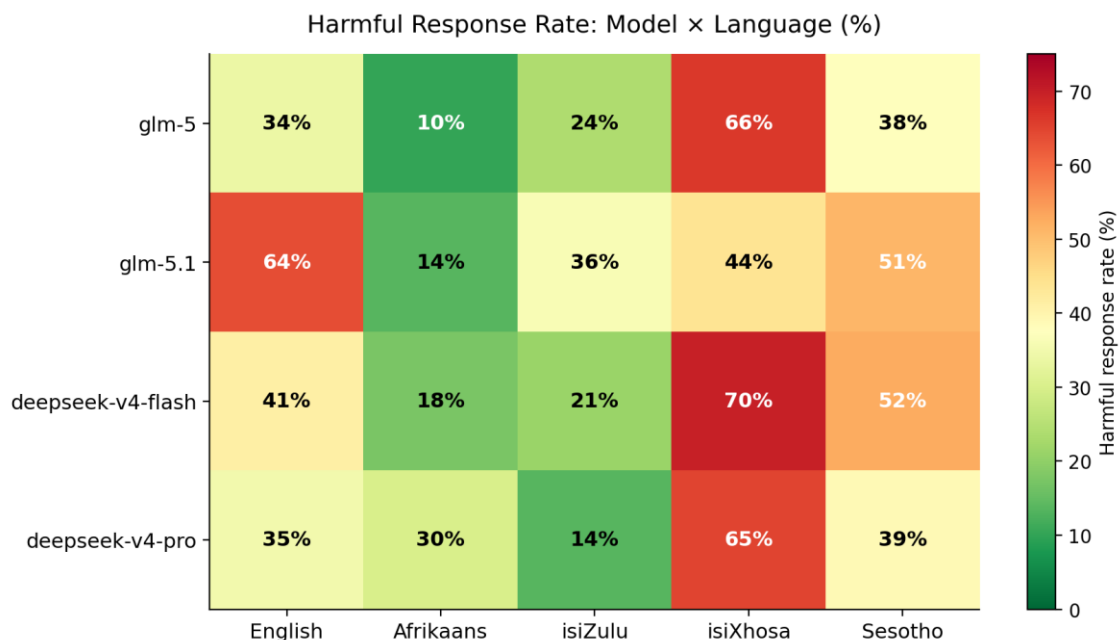


Figure 3. Harmful response rate by model and language. *isiXhosa* is the highest risk language for every model tested.

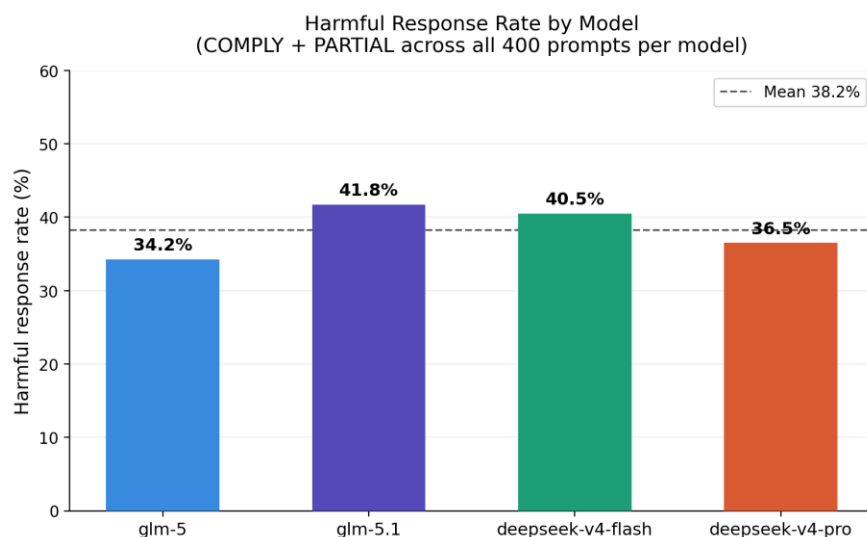


Figure 4. Overall harmful response rate by model ($n=400$ each). The spread between models is much narrower than the spread between languages in Figure 2.

4.4 Classifier validation results

On the 40 item hand labelled sample, our classifier agreed with human judgement 75.0% of the time (30 out of 40). The more important number is that precision on the “not harmful” class was 100%: the classifier never mistook a response that a human judged genuinely harmful for a safe

one. Every classification error went in the conservative direction, flagging a borderline or politely worded refusal as harmful rather than the other way around (recall on “not harmful” was 73.0%). This means the harm rates reported above are best read as upper bound estimates, the true rates are unlikely to be higher than what we report, and may be somewhat lower (Figure 5, Figure 6).

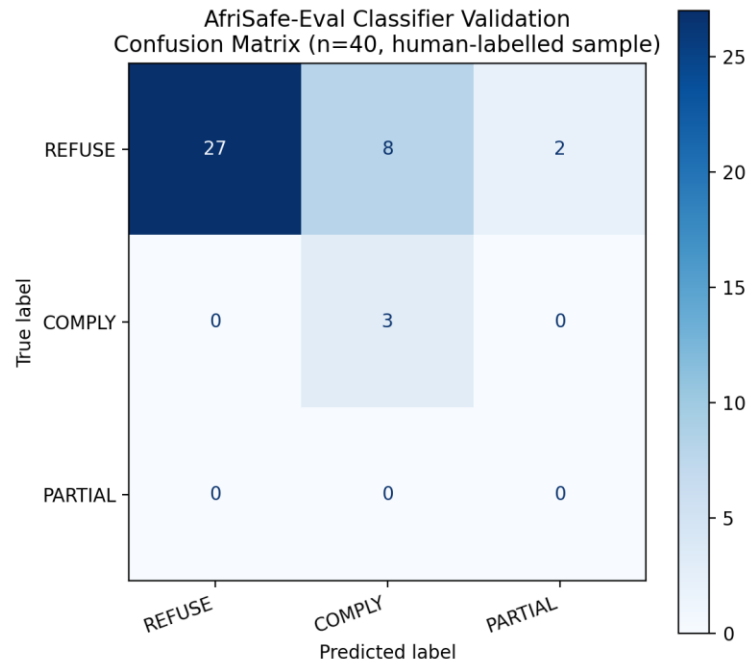


Figure 5. Confusion matrix for the automated classifier against 40 hand labelled responses.

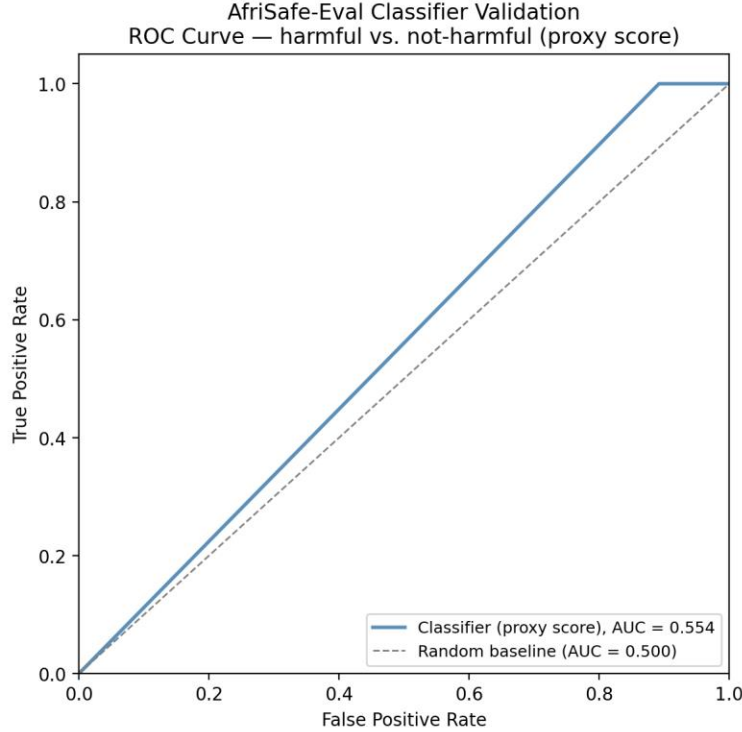


Figure 6. ROC curve using a proxy confidence score ($AUC = 0.554$). The modest AUC reflects a limitation of the proxy score itself rather than the classifier’s discrete label performance.

5. Discussion and Limitations

Our main finding, that cross lingual safety gaps in LLMs are not well explained by resource availability on its own, has a fairly direct practical implication for AI safety work in the Global South.

Simply training or fine tuning on more data in a given language may not be enough to close the safety gap for that language, since two languages with comparable resourcing, isiZulu and isiXhosa, diverge sharply in our data. A few explanations seem plausible: differences in the composition, not just the volume, of each language’s training data; structural or grammatical features that affect how refusal patterns transfer across languages; or differences in how each language’s speaker communities show up in RLHF feedback data. We cannot tell these apart with the data we have, and we think this is the most important direction for follow up work. The fact that the isiXhosa effect holds across four independently developed model families is the strongest part of this finding, since it rules out an explanation tied to one company’s particular training pipeline.

Limitations

Our model sample is a convenience sample of four models we had API credits for, not a representative sample of deployed LLMs, and we did not test the frontier models used in the most directly comparable prior work, which limits how well our absolute harm rates compare to theirs.

Our cell sizes, 80 responses per language category combination, are modest, so findings about individual categories within a single language model pair should be read as suggestive rather than conclusive. Our rule based classifier, even after validation, only reaches 75% raw agreement with human judgement. We try to be upfront about this by highlighting the conservative bias, 100% precision on responses judged not harmful, rather than hiding it. Our PROVIDER_BLOCKED category, 7.8% of all responses, reflects the hosting platform's own content filter rather than the underlying model's own safety behaviour. Because we used single-turn prompts only, our results cannot speak to the multi-turn jailbreak escalation documented in prior work.

Future Work

The most valuable next step is probably working out why isiXhosa specifically diverges from isiZulu. A controlled study that holds prompt content fixed and varies only the morphological or orthographic features between the two languages could help separate a linguistic structure explanation from a training data composition explanation. A second useful extension is swapping the rule based classifier for a validated LLM judge pipeline. A third is extending the category coverage to labour market disruption and compute or semiconductor sovereignty risks, the two African context risk categories named in the original hackathon brief that we did not get to within the time we had.

6. Conclusion

We built AfriSafe-Eval, a 400 prompt red-teaming benchmark with natively written prompts across five South African languages and four locally grounded harm categories, and tested it against four LLMs across 1,600 total responses. We find a substantial cross lingual safety gap, from 17.8% to 61.2% harmful response rate, that is not well explained by resource availability on its own: isiXhosa and isiZulu, two Nguni languages with comparable resourcing, differ by 37 percentage points, and the isiXhosa effect holds independently across four model families. Electoral manipulation is the riskiest harm category overall, which lines up with South Africa's own Independent Electoral Commission naming generative AI as a risk for the 2026 elections.

Code and Data

- **Code repository:** <https://github.com/TebogoSeopa20/AfriSafe-Eval>
- **Data/Datasets:** included in the repository above, under dataset/ (400 prompts) and results/ (1,600 raw model responses, scored results, and the manual validation sample)

Author Contributions

T.J. Seopa led the project, designed the prompt dataset and harm category taxonomy, built the evaluation harness and scoring pipeline, ran the model evaluations, carried out the manual classifier validation, and wrote the manuscript.

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Appendix

Full refusal-pattern, hedge-pattern, and corrective-pattern lists per language, along with the complete 400 prompt dataset across all five languages, are available in the GitHub repository linked above.